

Daniel Gedon

Postdoctoral Fellow in Machine Learning · Tübingen AI Center, University of Tübingen

✉ daniel.gedon@uni-tuebingen.de · 🌐 dgedon.github.io · 🐙 dgedon · 🎓 Google Scholar · 📄 dgedon

RESEARCH INTERESTS

Probabilistic machine learning, generative models, AI for scientific discovery, simulation-based inference, foundation models and LLMs, model discovery, uncertainty quantification.

TECHNICAL SKILLS

Languages & frameworks: Python (PyTorch, scikit-learn, numpy, pandas), Git

Infrastructure: SLURM, Hydra, Ray, Linux HPC clusters

Methods: Bayesian inference, simulation-based inference, generative models, sampling methods, transformers and LLMs (including pre-training from scratch), uncertainty quantification, hyperparameter optimization

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow, Tübingen University, Tübingen AI Center, Germany Sep 2024 – Present
Advisor: Jakob H. Macke

- Research on probabilistic ML and LLMs for AI-driven scientific discovery: LLM-based automated model discovery, and generative models and neural density estimation for simulation-based inference.
- Maintainer of `sbi` (800+ GitHub stars), a Python package for simulation-based inference.
- Student supervision, mentoring, and co-organization of SBI hackathons.

Satellite Attitude Control System Analyst, Airbus Defence & Space, Germany Oct 2015 – Sep 2016

- High-precision satellite pointing error analysis using ESA PEET simulation software.
- AOCS/GNC system concept design for early-phase satellite missions.
- Sensor modeling and functional simulation of attitude determination.

EDUCATION

Ph.D. in Machine Learning, Uppsala University, Sweden Aug 2019 – Aug 2024
Advisor: Thomas B. Schön · Thesis: *On Deep Learning for Low-Dimensional Representations* [Thesis]
Topics: deep learning for clinical ECG analysis, deep state-space models for system identification, self-supervised and kernel methods for representation learning
Visiting researcher, Belkin lab, UC San Diego, Spring 2023

M.Sc. in System and Control, TU Delft, Netherlands Sep 2017 – Jul 2019
Thesis: *Tensor Network Kalman Filter for Large-Scale MIMO Systems* [Thesis]

B.Eng. in Aerospace Engineering, DHBW, Germany Sep 2012 – Sep 2015

OPEN-SOURCE SOFTWARE

sbi 2024 – present
Python package for simulation-based inference. 800+ GitHub stars. Maintainer. [github]

GRANTS & AWARDS

Automatica Paper Prize Award, IFAC Aug 2026
Awarded at the 23rd IFAC World Congress, Busan, Korea, for *Deep Networks for System Identification: A Survey* (Automatica, 2025).

AI for Science Program, Anthropic Spring 2026
\$20,000 in Claude API credits for life science application

WASP Postdoctoral Fellowship , Knut and Alice Wallenberg Foundation Two-year scholarship at the Institute of Science and Technology Austria (ISTA). Awarded; declined.	Spring 2024
WASP Research Grant , Knut and Alice Wallenberg Foundation Funding for three-month research visit to Mikhail Belkin's lab, UC San Diego.	Spring 2023

ACADEMIC SERVICE

Organization

SBI Hackathon (5 days), Tübingen, Germany [website]	Mar 2025
SBI Hackathon/Tutorial (3 days), Grenoble, France [website]	Jan 2026
Sim2Science NeurIPS workshop, Paris, France [website]	Dec 2026

Peer Review

ML venues: NeurIPS (2023–2026), ICLR (2024–2025), ICML (2025), AISTATS (2023–2024), UAI (2025)
Other conferences: IFAC WC (2023), CDC (2023), ECC (2023), MIR (2024), IEEE CCTA (2023)
Journals: Automatica (2023), IEEE CSS (2025), IEEE TC-SMC (2023), IEEE TIM (2023), IEEE Access (2023), BMJ (2022)

SUPERVISION AND MENTORING

Shanzi Bao, student intern, Tübingen University	2026 – present
Johannes Schmidt, GSoC contributor (sbi), University of Bonn	Summer 2026
Anupam Sourav Patra, student assistant, Tübingen University	2026 – present
Ahmet Alperen Güngör, student assistant, Tübingen University	2026 – present
Stefan Wahl, PhD student, Tübingen University	2025 – present
Philipp von Bachmann, MSc project, Uppsala University	Spring 2022
Theogene Habineza, MSc thesis, Uppsala University	Spring 2022

TEACHING

Teacher , Tübingen University, Germany Seminar: Simulation-based inference (MSc) — Spring 2025 Teamproject: Benchmarking for Misspecified SBI (BSc) — Spring 2025 Seminar: Diffusion models (MSc) — Fall 2025	Sep 2024 – present
Lecturer , Uppsala University, Sweden Advanced Probabilistic Machine Learning, iRT705 (MSc), Fall 2023	Sep 2023 – Aug 2024
Teaching Assistant , Uppsala University & TU Delft Statistical ML, Advanced Probabilistic ML, System Identification, AI & ML for WASP PhD school — MSc/PhD level	Oct 2018 – Aug 2024

TALKS

<i>Simulation-based inference and probabilistic model discovery with foundation models</i> KTH, Stockholm	Jun 2026
<i>Simulation-based inference and probabilistic model discovery with foundation models</i> Chalmers, Gothenburg	Jun 2026
<i>Simulation-based inference and probabilistic model discovery with foundation models</i> Aiz, Seattle	Dec 2025
<i>Deep Networks for System Identification: A Survey</i> ERNSI Workshop, Stockholm	Sep 2023
<i>No Double Descent in PCA: Training and Pre-Training in High Dimensions</i> Belkin Lab Group Meeting, UC San Diego	Mar 2023
Panel: <i>Training in Data Driven Life Science</i> SciLifeLab DDLS Annual Conference, Stockholm	Nov 2022

OTHER EXPERIENCE

Solo Travel

Oct 2016 – Apr 2017

Long-distance hike, Greater Patagonian Trail, Patagonia. Language school: Spanish (Sucre, Bolivia).

Voluntary Work

Apr 2017 – Aug 2017

Work with primary school children, elderly, and refugees. Ansbach, Germany.

LANGUAGES

German (native) · English (fluent) · Swedish (intermediate) · Spanish (beginner)

PUBLICATIONS

* denotes equal contribution / joint supervision.

Preprints

Peer-Reviewed Publications

1. **Mixed neural posterior estimation for simulators with discrete and continuous parameters**
Jan Boelts, Cornelius Schröder, Jonas Beck, Jakob H. Macke*, Michael Deistler*, **Daniel Gedon***
accepted at TMLR, 2026 [arXiv] [OpenReview] [code]
2. **A Probabilistic Framework for LLM-Based Model Discovery**
Stefan Wahl, Raphaela Schenk, Ali Farnoud, Jakob H. Macke, **Daniel Gedon**
ICML, 2026 [arXiv] [OpenReview] [code]
3. **A Deep Learning ECG Model for Localization of Occlusion Myocardial Infarction**
Stefan Gustafsson, Antônio H. Ribeiro, **Daniel Gedon**, Petrus E.O.G.B. Abreu, Nicolas Pielawski, Gabriela M.M. Paixão, Antonio Luiz P. Ribeiro, Daniel Lindholm, Thomas B. Schön, Johan Sundström
Nature Communications, 2026 [DOI] [code]
4. **Effortless, Simulation-Efficient Bayesian Inference using Tabular Foundation Models**
Julius Vetter, Manuel Gloeckler, **Daniel Gedon***, Jakob H. Macke*
NeurIPS, 2025 [arXiv] [OpenReview] [code]
5. **Deep Networks for System Identification: A Survey**
Gianluigi Pillonetto, Aleksandr Aravkin, **Daniel Gedon**, Lennart Ljung, Antônio H. Ribeiro, Thomas B. Schön
Automatica, 2025 [DOI] [arXiv]
🏆 Automatica Paper Prize Award 2026
6. **No Double Descent in Principal Component Regression: A High-Dimensional Analysis**
Daniel Gedon, Antônio H. Ribeiro, Thomas B. Schön
ICML, 2024 [OpenReview] [ICML] [code]
7. **Uncertainty Estimation with Recursive Feature Machines**
Daniel Gedon, Amirhesam Abedsoltan, Thomas B. Schön, Mikhail Belkin
UAI, 2024 [OpenReview] [code]
8. **Evaluating Regression and Probabilistic Methods for ECG-based Electrolyte Prediction**
Philipp Von Bachmann, **Daniel Gedon**, Fredrik K. Gustafsson, Antônio H. Ribeiro, Erik Lampa, Stefan Gustafsson, Johan Sundström, Thomas B. Schön
Scientific Reports 14, 15273, 2024 [DOI] [arXiv] [code]
9. **End-to-end Risk Prediction of Atrial Fibrillation from the 12-Lead ECG by Deep Neural Networks**
Theogene Habineza, Antônio H. Ribeiro, **Daniel Gedon**, Joachim A. Behar, Antonio Luiz P. Ribeiro, Thomas

- B. Schön
Journal of Electrocardiology, 2023 [DOI] [arXiv] [code]
10. **Screening for Chagas Disease from the Electrocardiogram using a Deep Neural Network**
 Carl Jidling, **Daniel Gedon**, Thomas B. Schön, Claudia Di Lorenzo Oliveira, Clareci Silva Cardoso, Ariela Mota Ferreira, Luana Giatti, Sandhi Maria Barreto, Ester C. Sabino, Antonio L. P. Ribeiro, Antônio H. Ribeiro
PLOS Neglected Tropical Diseases, 2023 [DOI] [medRxiv] [code]
 11. **Invertible Kernel PCA with Random Fourier Features**
Daniel Gedon, Antônio H. Ribeiro, Niklas Wahlström, Thomas B. Schön
IEEE Signal Processing Letters, 2023 [DOI] [arXiv] [code]
 12. **Development and Validation of Deep Learning ECG-based Prediction of Myocardial Infarction in Emergency Department Patients**
Daniel Gedon^{*}, Stefan Gustafsson^{*}, Erik Lampa, Antônio H. Ribeiro, Martin J. Holzmänn, Thomas B. Schön, Johan Sundström
Scientific Reports 12, 19615, 2022 [DOI]
 13. **First Steps Towards Self-Supervised Pretraining of the 12-Lead ECG**
Daniel Gedon, Antônio H. Ribeiro, Niklas Wahlström, Thomas B. Schön
Computing in Cardiology (CinC), 2021 [DOI] [slides] [video]
 14. **Deep State Space Models for Nonlinear System Identification**
Daniel Gedon, Niklas Wahlström, Thomas B. Schön, Lennart Ljung
Proceedings of the 19th IFAC Symposium on System Identification (SYSID), 2021 [DOI] [arXiv] [code]
 15. **Automatic 12-Lead ECG Classification using a Convolutional Network Ensemble**
 Antônio H. Ribeiro, **Daniel Gedon**, Daniel Martins Teixeira, Manoel H. Ribeiro, Antonio L. Pinho Ribeiro, Thomas B. Schön, Wagner Meira Jr.
Computing in Cardiology (CinC), 2020 [DOI] [code]
 16. **Tensor Network Kalman Filter for LTI Systems**
Daniel Gedon, Pieter Piscaer, Kim Batselier, Carlos Smith, Michel Verhaegen
27th European Signal Processing Conference (EUSIPCO), A Coruña, Spain, 2019 [DOI] [code]
 17. **PointingSat – High Precision Pointing Error Analysis with ESA PEET v1.0**
 Thomas Ott, Marc Hirth, Massimo Casasco, Simon Görries, **Daniel Gedon**, Alison Ponche
10th International ESA Conference on Guidance, Navigation & Control Systems, Salzburg, Austria, 2017 [Paper]